AZISLY

AI Career Readiness Platform

Sprint QA Internship Assignment

Testing Strategy · AI Output Quality Framework · QA Prototype

Prepared by: Abhay Kumar | Role: AI-Native Quality Analyst

Sprint: College Dashboard · Interview Prep v2 · API /api/user/profile Update

Platform: Web · Android · iOS

Deliverable 1 – Testing Strategy Document

ℹ This document covers the full QA plan for the current sprint: College Dashboard (new), Interview Prep Module (LLM upgrade), and the /api/user/profile API change. All strategy decisions are grounded in Azisly's actual product context.

1.1 Sprint Coverage Map

The table below maps every sprint item to its required testing types. A filled cell means that testing type is required; an empty cell means it does not apply.

Sprint Item	Functional	Security / Access Ctrl	AI Output Quality	Data Correctness	Cross-Platform	Regression
College Dashboard	✓ Core role flows	✓ CRITICAL – privacy	—	✓ Aggregation logic	✓ Web only	✓ Login system
Interview Prep Module	✓ Prompt triggers	✓ Data in prompts	✓ CRITICAL – LLM	✓ Role-type routing	✓ Web + Mobile	✓ Old prompt removal
API /user/profile	✓ CRUD new fields	✓ Auth, field exposure	—	✓ Field storage/sync	✓ All 3 platforms	✓ Existing fields safe

1.2 Manual vs Automation Decisions

The reasoning column is the most important part — blanket automation or blanket manual testing is a red flag in any serious QA context.

Test Area	Approach	Sprint Item	Reasoning
College cannot view individual student data	**Manual (exploratory)**	College Dashboard	This is a privacy boundary. Automated checks can verify status codes, but a real tester must attempt data extraction via unexpected UI paths, filter manipulation, and direct API calls to catch logic gaps the test author didn't anticipate.
Anonymous aggregation accuracy	**Automated (data layer)**	College Dashboard	Aggregated counts are deterministic — given a seeded dataset, the expected dashboard values are known. Automating this removes human counting error and runs fast on every build.
Interview Prep – role-specific response routing (technical vs non-technical)	**Manual + LLM eval harness**	Interview Prep Module	Routing logic (which prompt is called) can be automated. But whether the LLM response is actually role-appropriate requires human judgment or a secondary LLM evaluator — you cannot assert on free text with a simple string match.
Old prompt system is fully removed	**Automated (code search + smoke)**	Interview Prep Module	A grep/AST scan of the codebase for old prompt identifiers + a smoke test confirming the old endpoint/path no longer exists is sufficient and fast.

API – career_goal / target_role field write and read	**Automated (contract tests)**	/api/user/profile	These fields have deterministic inputs and outputs. Contract tests (Postman collection or Pact) should verify field presence, correct data types, and that missing fields return appropriate defaults — not 500 errors.
Cross-platform rendering of new profile fields	**Manual (device matrix)**	/api/user/profile	No API versioning means web and both mobile apps consume the same response simultaneously. A manual spot-check on iOS and Android is needed because mobile apps may not have been updated to handle new fields gracefully — especially if old app versions are still in the wild.
LLM response factual accuracy for domain-specific advice	**Manual (QA + SME review)**	Interview Prep Module	No automated test can reliably determine if interview advice for a 'Data Engineer at a Series B startup' is factually correct without domain knowledge. This requires a human reviewer or curated golden set.

1.3 Risk Analysis Table

Risk Name	Sprint Item	Test Approach	Impact if Missed
R1 – College accesses individual student record CRITICAL	College Dashboard	Manual exploratory + API auth tests	FERPA / privacy violation. Institutional trust destroyed. Students may drop platform. Legal exposure for Azisly.
R2 – LLM gives confidently wrong interview advice CRITICAL	Interview Prep Module	Manual review + output eval harness	Student acts on bad advice in a real interview. Core product value destroyed. Negative reviews. Churn.
R3 – No API versioning breaks mobile apps HIGH	/api/user/profile	Manual device test + contract test	Old mobile app versions crash or silently discard career_goal / target_role. Feature invisible to large portion of mobile users who haven't updated.
R4 – Role routing sends wrong prompt type (tech vs non-tech)	Interview Prep Module	Automated routing unit test + manual review	Non-technical student gets Python DSA question. Technical student gets generic behavioral response. Personalization — the core differentiator — breaks.
R5 – Aggregated college dashboard counts are wrong	College Dashboard	Automated data tests with seeded DB	College partners receive inaccurate engagement reports. Institutional relationships at risk.
R6 – Old prompt artifacts still reachable in production	Interview Prep Module	Codebase scan + smoke test	Inconsistent student experience. Possibly insecure if old prompts had looser guardrails.
R7 – career_goal persists to wrong user account	/api/user/profile	Auth boundary test (automated)	User data privacy breach. One student sees another's career goal. Trust violation.

1.4 Top 3 Prioritized Risks — Justification

Risk 1 (R1): College Accesses Individual Student Data — CRITICAL

> This is the #1 risk for this sprint. The College Dashboard is brand-new, and the privacy boundary between aggregated analytics and individual records is entirely enforced by Azisly's own code — there is no external system guaranteeing it. A single misconfigured API permission or a dashboard filter that drills down too deep is a FERPA violation. This cannot be caught post-release; by the time a college admin discovers they CAN see individual data, the damage is done.

Risk 2 (R2): LLM Gives Confidently Wrong Interview Advice — CRITICAL

> The entire Interview Prep Module has been re-architected. The new LLM prompt system has never been in production. Unlike UI bugs that are visually obvious, a confidently wrong LLM response looks completely legitimate to a student who trusts the product. A student who bombs an interview after following Azisly's advice — and realizes the advice was wrong — is a churned user and a public reputation risk.

Risk 3 (R3): No API Versioning Breaks Mobile — HIGH

> This sprint adds two new fields to /api/user/profile with no versioning strategy. The assumption that all mobile clients will handle unexpected new fields gracefully is wrong — it depends on how each mobile app was coded. If the Android or iOS client deserializes the profile object into a strict model, an unknown field can throw a parse exception. Given that mobile app store updates lag by days, there will always be users on the old version when the API goes live.

1.5 4-Hour Release Constraint: Testing Sequence

> ℹ Assumption: The 4 hours begins after a successful build deployment to staging. Two QA testers available. Web and one physical device (iOS or Android) accessible.

Block	Time	What to Test	Why This Slot / Skip Rationale
1	0:00 – 0:45	College Dashboard: privacy boundary. Attempt to access individual student data via UI, direct API calls, filter manipulation, and URL enumeration. Verify role-based access blocks fire correctly.	R1 is the highest-stakes risk. If this fails, release must be blocked regardless of anything else. Do this first, while testers are fresh.
2	0:45 – 1:30	Interview Prep Module: run 6–8 representative prompts across technical and non-technical roles. Check role routing fires correctly. Spot-check one response per type for factual plausibility.	R2. New LLM architecture has never been production-tested. Routing failure would mean the entire module is broken for a user class.
3	1:30 – 2:00	/api/user/profile: POST and GET with career_goal and target_role on web. Verify fields persist correctly and existing fields are untouched. Run contract test collection.	R3/R7. API contract break is fast to detect and blocks the most users (web + mobile).
4	2:00 – 2:30	Mobile spot-check (one device): load updated profile, verify new fields display without crash. Attempt career_goal update from mobile.	No versioning means mobile is at risk. Even a 15-minute smoke test on one device catches a hard crash scenario.

5	2:30 – 3:15	College Dashboard: verify aggregation counts against seeded test data. Check dashboard renders correctly on web. Test college login flow end-to-end.	R5. Functional correctness of the dashboard — still critical but lower urgency than privacy boundary.
6	3:15 – 3:45	Regression: verify existing student login, existing interview prep module entry points, and existing profile fields are unaffected.	Three sprint items all touch shared systems. A quick regression pass on non-changed flows takes 30 minutes and catches unexpected breaks.
Buffer	3:45 – 4:00	Document defects found. Go/No-Go decision.	Any blocking defect found in Blocks 1–2 = No-Go. Blocks 3–6 defects = risk-rated go/no-go with PM.

What to Skip Under 4-Hour Constraint

- Full cross-platform UI testing across all 3 platforms — defer iOS + Android full regression to post-release monitoring.
- Load testing or performance benchmarking on the College Dashboard.
- Edge cases in LLM prompt behavior beyond basic role routing (those belong in ongoing AI quality monitoring, not a release gate).
- Visual/design review of the new dashboard — flag to PM, not a release blocker.

Deliverable 2 – AI Output Quality Framework

ℹ This framework is purpose-built for Azisly's interview coaching outputs — not a generic LLM eval rubric. Every dimension and failure mode is grounded in the real-world context of a student using AI to prepare for a job interview.

2.1 Quality Dimensions

A strong Azisly coaching response must satisfy all five dimensions. A failure on any dimension degrades the student's preparation.

Dimension	What It Means for Azisly	Test Signal
Role Relevance	The response must be calibrated to the student's declared role type (technical vs non-technical) and target domain. A software engineering student and a marketing student asking 'how do I answer tell me about yourself' need structurally different answers.	Compare response against a role-type rubric. Presence of irrelevant technical jargon in a non-tech response = fail.
Factual Accuracy	Any specific claim about interview processes, company practices, or technical knowledge must be correct. 'Amazon uses the STAR method' is a factual claim. 'System design interviews are typically 30 minutes' may be incorrect for many companies.	Cross-reference against known interview process documentation. Domain SME spot-check.
Actionability	The student should be able to do something with the response immediately. Vague affirmations ('great question, interviews can be challenging!') add zero value to a student who has an interview in 48 hours.	Count actionable instructions vs filler sentences. Ratio below 50% actionable = flag.
Appropriate Confidence	The model must signal uncertainty when it exists. Presenting speculative or variable information as definitive fact ('Google always asks this question') is actively harmful.	Flag declarative statements about variable/unknowable facts without hedging language.
Personalization	Azisly knows the student's career_goal and target_role (new this sprint). The coaching response should reflect this context. A response that could have been given to any student regardless of their profile is a failure of the personalization promise.	Verify new career_goal / target_role fields appear to influence the response. A/B same question with different profiles — responses must diverge meaningfully.

2.2 Failure Taxonomy

Failure Type	Azisly-Specific Example	Real-World Student Impact
Factually Incorrect	Model states: 'For data analyst roles, you'll typically face a take-home case study of 5–7 days.' In reality, most entry-level analyst screens are 1–2 day take-homes or synchronous SQL tests.	Student under-prepares for speed, thinking they have a week. Fails the screen. Blames themselves, not the tool.

Contextually Wrong	Student profile: 'target_role: UX Designer.' Question: 'How should I prep for a technical screen?' Model recommends LeetCode and system design prep — technically coherent, completely wrong context.	Student wastes 10+ hours on irrelevant prep. Enters the actual design portfolio review completely unprepared.
Generic / Not Personalized	Student has career_goal: 'transition from teaching to product management.' Model gives a standard 'how to answer behavioral questions' response with no acknowledgment of the career pivot narrative, which is the student's biggest challenge.	Student receives no value that they couldn't get from any free resource. Platform differentiation evaporates. Churn risk.
Confidently Wrong	'Stripe's engineering interviews always include a distributed systems design problem in round 2.' Stated as fact. Stripe's interview format changes by team and has evolved over time.	Student over-indexes on distributed systems prep for a company that might give them a product sense or frontend-focused interview. High-stakes error delivered with authority.

2.3 Subtle Edge Cases

Edge Case 1 – The Correct-But-Outdated Response

A student asks about interview prep for a company that recently changed its process (e.g., many tech companies eliminated LeetCode-style screens post-2023 and shifted to take-home projects). The model, trained on older data, gives advice that was accurate 18 months ago but is now misleading. The response passes a surface-level factual check because it was once true.

> ⚠ This is especially dangerous post-LLM architecture change, since the new prompt system may be pulling from training data with a knowledge cutoff that predates significant industry shifts.

Edge Case 2 – Role-Type Routing Ambiguity

A student declares target_role: 'Data Scientist' and career_goal: 'move into ML engineering.' The routing logic must decide: is this a technical or non-technical profile? The answer is clearly technical — but the ambiguity arises when the student asks a behavioral question like 'how do I talk about my projects?' A non-technical prompt template might produce generic advice, while the technically-routed prompt would correctly guide them on communicating ML model performance to non-technical stakeholders.

> ⚠ The new LLM prompt architecture's handling of hybrid-role profiles has not been defined in the sprint spec. Assuming the routing is binary; if nuanced routing exists, additional test cases are needed.

Edge Case 3 – The Personalization Illusion

The model mentions the student's target_role in the first sentence ('As someone targeting a Product Manager role...') and then gives entirely generic advice. The response feels personalized because it name-checks the role, but the body of the response is identical to what any student would receive. This passes a naive 'does the response mention the role' check but fails the actual personalization quality bar.

> ℹ Testing implication: personalization checks must evaluate the body of the response, not just presence of keywords. A/B the same question across two very different profiles — if the middle 80% of the response is identical, it is not personalized.

2.4 Hypothetical Coaching Responses — Pass vs Fail

Student Context

Name: Priya | target_role: Business Analyst | career_goal: First job after MBA | Question: 'How should I prep for a case interview?'

Response A — PASS ✓

> *For a Business Analyst role right out of your MBA, case interviews are typically structured problem-solving exercises rather than full McKinsey-style strategy cases — though this varies by company type. For consulting-adjacent BA roles, practice structuring your thinking out loud using a framework like Issue Tree or MECE. For corporate BA roles (think fintech, healthcare ops), you're more likely to face a data interpretation exercise or a 'walk me through a process improvement you'd suggest' scenario based on a provided dataset. I'd recommend identifying 2–3 companies on your target list and researching their specific BA interview format — LinkedIn and Glassdoor reviews from the last 6 months are your best signal. Once you know the format, come back and we can work through specific practice prompts together.*

- Role-relevant: distinguishes consulting-adjacent vs corporate BA. ✓
- Actionable: gives specific research steps and next action. ✓
- Appropriately confident: hedges on company variation ('varies by company type'). ✓
- Personalized: acknowledges MBA context which shapes the types of cases she'll face. ✓
- No false certainty about specific company formats. ✓

Response B — FAIL ✗ (Subtle)

> *Great question! Case interviews are a common part of the Business Analyst hiring process. To prepare, you should practice breaking down problems into structured components and communicating your reasoning clearly. Focus on frameworks like SWOT analysis and market sizing. It's also helpful to practice mental math and work through common business scenarios. Most importantly, show your ability to think analytically and communicate effectively. Good luck with your interviews!*

Why this FAILS (subtle — it reads professional at first glance):

- Generic: Every word of this could be given to any student asking any career question. Priya's MBA, her BA target, and her career stage are completely ignored.
- Low actionability: 'practice breaking down problems' and 'communicate effectively' are not instructions. There is nothing Priya can do tonight.
- Factually shallow: SWOT is not a standard case interview framework. This conflates business school coursework with case interview prep.
- Confidently generic: 'Case interviews are a common part of the BA hiring process' — this is only true for certain BA roles. Many corporate BA interviews have no case component at all.
- No personalization signal: zero use of career_goal ('first job after MBA') which is the defining context for how she should frame her experience.

Deliverable 3 – Testing Prototype Sketch

> ℹ This prototype is designed specifically for this sprint. It is not a general QA tool — it addresses the highest-risk gap identified in Deliverable 1: the lack of any structured process to evaluate whether the new LLM prompt architecture is producing role-appropriate, non-harmful coaching outputs at release time.

3.1 Prototype Name

Azisly Interview Coaching Output Evaluator (AICOE) — Sprint Smoke Test Edition

3.2 What It Tests

AICOE is a lightweight, semi-automated evaluation harness that runs a curated set of student prompt scenarios through the live Interview Prep Module and scores the outputs against the quality dimensions defined in Deliverable 2. It is designed to run in under 45 minutes as part of the pre-release checklist.

3.3 Inputs

Input	Description	Source
Student Persona Set	8–12 synthetic student profiles with varying target_role, career_goal, and experience level. Curated to cover: tech / non-tech routing, career transition scenarios, fresh graduates vs experienced switchers.	QA team creates & maintains in a shared JSON file.
Prompt Scenario Library	15–20 standard interview prep questions mapped to persona type. Examples: 'How do I answer Tell me about yourself?', 'What should I expect in a system design round?', 'How do I address a career gap?'	QA + Product team. Reviewed each sprint.
Evaluation Rubric (Scorecard)	5 binary (pass/fail) + 2 scaled (1–3) criteria per response. Defined in Deliverable 2. Binary: Role Relevance, No False Certainty, Actionable Step Present, No Generic Filler Opening, Personalization Signal. Scaled: Factual Plausibility (1–3), Actionability Depth (1–3).	QA document — updated when failure taxonomy expands.
Previous Sprint Baseline	Archived scored outputs from the prior sprint for the same persona/prompt pairs. Used for regression comparison.	AICOE output archive (CSV).

3.4 Logic / Process Flow

Step	Action	Details
1	Seed test profiles	Load 8–12 synthetic student profiles into the staging environment via the /api/user/profile endpoint, including the new career_goal and target_role fields. Verify each profile is correctly stored before proceeding.
2	Run prompt scenarios	For each persona × prompt pair (a subset: ~20 combinations for the smoke test), submit the question to the Interview Prep Module as that student. Log the raw response, response latency, and which prompt template was invoked (if loggable).

3	Score each output	Two reviewers independently score each response against the 7-criteria rubric. Binary criteria are marked Pass/Fail. Scaled criteria are marked 1–3. Disagreements on binary criteria are escalated; scaled criteria use the lower score (conservative).
4	Flag failures	Any response with: 2+ binary fails, OR a Factual Plausibility score of 1, OR a Confidently Wrong pattern (declarative + unverifiable) is marked a DEFECT and logged with the full response text, persona context, and criteria breakdown.
5	Regression diff	Compare current sprint scores against the baseline archive. Any persona/prompt pair that previously passed and now fails = regression defect. Flag immediately to engineering.
6	Generate report	Output a one-page CSV + summary (see 3.5 below). Attach to the Go/No-Go decision document.

3.5 Output / Results Format

Output	Format & Content
Per-Response Scorecard	CSV row: Persona ID | Prompt ID | Role Type | Response (truncated 200 chars) | Binary Criteria (5 cols: P/F) | Factual Score | Actionability Score | Overall Status (Pass / Flag / Defect) | Reviewer 1 | Reviewer 2
Sprint Summary	One-page summary: total scenarios run, % pass rate by role type (tech vs non-tech), defects found, regression defects, top failure mode this sprint. Includes a one-sentence risk statement for the Go/No-Go meeting.
Defect Log	Separate tab: full response text, failure reason, severity (Critical / High / Medium), suggested prompt fix hypothesis, assigned to: [engineer / prompt designer].
Trend Baseline Update	If sprint passes, AICOE automatically promotes current outputs to the new baseline archive for the next sprint comparison.

3.6 Scope Limitations

- AICOE tests coaching output quality — it does not test the interview prep module's UI, navigation, or performance.
- The rubric relies on human reviewers for factual accuracy scoring. AICOE does not automate factual verification (a secondary LLM evaluator could be added in a future sprint).
- The persona set covers the most common user archetypes but cannot cover all possible target_role × career_goal combinations. Edge case personas are added iteratively based on real user feedback.
- AICOE is a smoke test, not a comprehensive quality audit. It is designed to catch blocking defects before release — not replace ongoing AI quality monitoring.
- Currently scoped for web platform only. Mobile-specific prompt behavior (if it differs) is not covered in v1 of this prototype.

3.7 Why This Prototype Matters for This Sprint

This sprint replaces the entire LLM prompt architecture for Interview Prep. The old system is removed. There is no safety net of a 'known good' prompt system to fall back on — once this ships, every student interaction goes through the new system.

> ✅ Without AICOE, the only way to catch a prompt regression is through production user complaints — which means real students, in real interview prep sessions, receiving degraded or harmful advice before anyone at Azisly knows something is wrong. AICOE gives the team a structured, documented, repeatable quality gate that takes 45 minutes and produces a release-ready artifact.

Additionally: the new career_goal and target_role fields from the API change are designed to feed directly into personalization. AICOE is the only test that verifies this integration actually works end-to-end — that the new fields influence the coaching response in a meaningful way, not just at the data layer.

End of Internship Assignment Submission

Prepared by: Abhay Kumar

All assumptions are marked explicitly. All strategy decisions include reasoning. Designed for Azisly's specific sprint — not generic QA templates.